

FLAVOURBENCH: Executable Culinary Reward Maps for Language Model Evaluation and Post-Training

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Abstract

Open-ended language-model evaluation often substitutes another model or a small preference panel for a missing answer key. We introduce FLAVOURBENCH, which instead compiles dense answer maps from a versioned culinary environment. Each task asks for three ingredients from eight; before inference, EPICURE scores all 56 legal portfolios. We evaluate 27 frontier endpoints on the same 534 substitution, pairing, and constraint tasks, yielding 14418 complete observations. Anchor-cluster bootstraps and Holm-controlled paired tests resolve 101 of 351 model contrasts. Grok 4.6 has the largest point estimate (65.1), but corrected evidence does not identify a unique best endpoint. Rankings replicate across independently compiled panels and remain similar under alternative metrics, task filters, family weights, and three public EPICURE checkpoints. Those checkpoints also rank human-observed Recipe1MSubs targets above within-food-group chance. We then run a preregistered, three-seed post-training study. LoRA SFT of a pinned Qwen3-0.6B checkpoint on 270 EPICURE-optimal answers improves its score on 84 anchor-disjoint maps by 13.30 points over a format- and label-matched control (95% CI 6.52–20.29, $p = 1.70 \times 10^{-4}$), and the effect replicates on all 534 public maps (+11.73 points, 95% CI 8.98–14.54). On the primary split, both trained arms parse every response while format training alone does not improve on the base. The release contains prompts, reward maps, raw responses, routes, training and evaluation manifests, code, and offline verifiers.

1 Introduction

Benchmarks are strongest when the evaluator can be executed. Code can be tested, database states can be inspected, and games can be scored. Culinary reasoning is usually evaluated with a model judge, a handful of human preferences, or factual questions whose answers do not capture the quality of a complete decision. That leaves two problems: the judge is entangled with the systems being tested, and exact match throws away useful differences between plausible answers.

FLAVOURBENCH turns a culinary runtime into a benchmark environment. EPICURE represents 1,790 ingredients in a 300-dimensional space and exposes deterministic operations for substitution, pairing, dietary feasibility, and regional composition [15, 16]. A task compiler converts these operations into three-of-eight selection problems. It enumerates every candidate portfolio and freezes a continuous 0–100 score map before model execution. An answer is therefore scored by table lookup, without a model judge or interpretation after the fact.

EPICURE is not a leaderboard entry and is not “100% accurate.” It defines the task-specific reward surface. A model receives 100 on a task only if it selects that task’s highest-scoring portfolio; other valid portfolios receive partial credit. The benchmark measures agreement with a released culinary reward map, not universal human taste.

We use that environment to evaluate 27 current endpoints. Every model faces the same 534 tasks and 56 frozen answer scores per task. The complete common

core supports direct paired comparisons on 534 shared observations per model pair instead of inferring a ranking from sparse votes or comparing endpoints on different task subsets.

The paper makes four contributions:

1. an executable culinary benchmark with 29904 model-independent portfolio scores and a single interpretable 0–100 metric;
2. a 27-model, 534-task complete-core evaluation with simultaneous uncertainty and paired, multiplicity-controlled tests;
3. post-collection tests of task-filter, metric, family-weight, and reward-map dependence, including a rescore against three public EPICURE checkpoints and a held-out, human-observed substitution check; and
4. a preregistered three-seed study showing that EPICURE-optimal SFT transfers to unseen reward maps beyond format learning, backed by a public-map replication and content-addressed release.

2 Executable culinary reference environment

2.1 A reference environment, not a model judge

The primary task maps bind one exact EPICURE data bundle and application build by hash. The data and application bytes are public at an immutable source revision, so the released score maps can be reconstructed exactly.

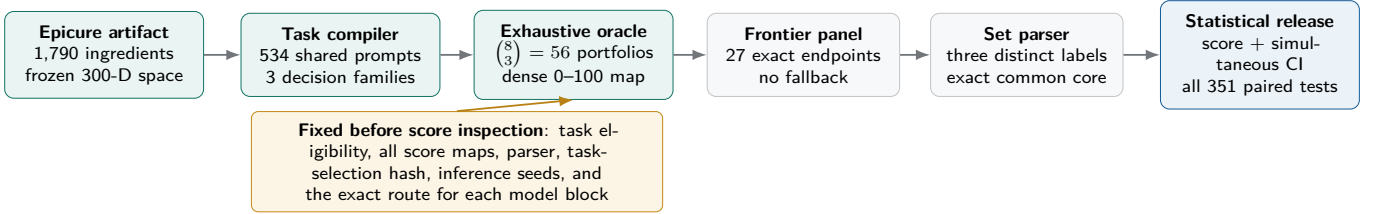


Figure 1: FLAVOURBENCH compiles a content-addressed culinary artifact into dense selection tasks. The evaluated models never judge one another and do not access EPICURE. All score maps and the per-panel contracts are fixed before collection. The complete-core task set is selected from response validity alone, before inspecting any model selection or score.

The original training run, seed, and source revision for this runtime were not recovered, and the artifact is not identified as the published Cooc, Core, or Chem checkpoint. We therefore use it as a fixed reference function, not as independently established culinary truth. Changing the artifact creates a new task-set identity. A separate post-hoc analysis below replaces its reward maps with the three public, paper-linked EPICURE checkpoints.

For each task t , let $\mathcal{A}_t$ contain all 56 three-item subsets of eight candidates. EPICURE assigns a raw utility $u_t(a)$ to every valid $a \in \mathcal{A}_t$. Invalid portfolios under a dietary or category constraint receive zero. Valid utilities are normalized within the task:

$$s_t(a) = 100 \frac{u_t(a) - \min_{b \in \mathcal{A}_t} u_t(b)}{\max_{b \in \mathcal{A}_t} u_t(b) - \min_{b \in \mathcal{A}_t} u_t(b)}. \quad (1)$$

Every task has a unique 100-point optimum. The exact chance baseline is not a generic percentage; it is $|\mathcal{A}_t|^{-1} \sum_a s_t(a)$ for that frozen task.

2.2 Three complete-core decision families

The ranked core contains 534 tasks: 89 from each family in each of two independently compiled panels, spanning 534 unique anchor ingredients. Tasks sharing an anchor move together as one cluster in inference. Candidate sets are selected across validation strata, then ordered and labeled by a task-derived hash. No evaluated model writes an item, distractor, or score. Regional-composition tasks are released as a supplemental track but are not in the 27-model leaderboard because they do not support an equally complete common core.

The score maps have substantial resolution rather than one correct key plus 55 equivalent mistakes. Table 2 reports the exact random-portfolio baseline, separation of the best and second-best portfolios, and the number of distinct score values. Constraint tasks are intentionally sparse because infeasible portfolios score zero; the other families provide much finer graded rewards.

2.3 Prompt and parser contract

Prompts show the decision context and eight labeled candidates. The final line must begin with the

FINAL_SELECTION marker and contain three distinct A–H labels separated by commas. No prompt supplies a concrete valid triplet. The parser takes the final exact marker line, uppercases the labels, sorts them as an unordered set, and performs one lookup in the frozen score map. Explanatory text does not affect the result. Missing markers, duplicate labels, provider failures, and abnormal finishes remain in the raw response ledger but cannot enter the common core.

The model-only condition is deliberate. The fixed reward map is not exposed as a tool during ranking. The score therefore measures whether a model can make the culinary decision, rather than whether it can copy a runtime return value after being told which operation to call.

3 Evaluation design

3.1 One primary score

For model m , panel p , family f , and task t , let Y_{mt} be the frozen portfolio score and let T_{pf} be the score-blindly selected set of 89 tasks completed by every model. The FlavourBench Score is

$$F_m = \frac{1}{2 \cdot 3} \sum_{p=1}^2 \sum_{f=1}^3 \frac{1}{|T_{pf}|} \sum_{t \in T_{pf}} Y_{mt}. \quad (2)$$

It ranges from 0 to 100 and equals the ordinary mean over all 534 cells because the six panel–family strata have equal size. Every ranked endpoint contributes exactly 534 valid responses. Failures are neither assigned zero nor silently dropped model by model: they determine whether a task can enter the shared core, using response status and parser validity only. The task-selection hash is fixed before loading any selected portfolio or score.

3.2 Frontier endpoint panel

The panel spans OpenAI, Anthropic, Google, xAI, Meta, Kimi, Qwen, Z.ai, DeepSeek, ByteDance, Thinking Machines, MiniMax, NVIDIA, Mistral, Tencent, and Cohere. It includes GPT-5.6 Sol, Terra, and Luna; Claude Opus, Sonnet, and Fable 5; Gemini 3.1 Pro and 3.6 Flash; Grok

Family	Decision	Epicure utility	Hard constraint	Tasks
Substitution	choose a three-item replacement portfolio for an anchor ingredient	0.8 anchor similarity +0.2 portfolio coherence	none	178
Pairing	choose three ingredients to accompany an anchor	0.65 anchor affinity +0.35 portfolio coherence	none	178
Constraints	choose a feasible portfolio under diet and processing limits	0.7 anchor similarity +0.3 coherence	diet and maximum NOVA level	178

Table 1: The three complete-core task families share one response and scoring interface but probe different culinary decisions. Each task contains eight candidates and 56 frozen scores.

Family	Exact chance	Top gap	Distinct scores
Substitution	45.2	5.6	56
Pairing	45.0	5.6	56
Constraints	3.4	37.3	4

Table 2: Frozen task-map diagnostics for 178 scheduled tasks per family, computed before model execution. Chance is the uniformly random portfolio mean; top gap is the median best-second-best margin.

4.6; Llama 4 Maverick; Muse Spark 1.2 and Muse Glimmer 30B; Kimi K3; Qwen 3.8 Max and Qwen3.8 2.4T A95B; GLM 5.2 and GLM 5.3; DeepSeek V4 Pro 0813 and V4 Flash; Seed 2.1 Turbo; Inkling; MiniMax M3; Nemotron 3.5 Lightning; Mistral Large 3; Tencent HY 3; Cohere Command A; and Cohere Command R+ (08-2024). Z.ai supplied written permission for one finite GLM 5.3 benchmark run, conditioned on the adapter not becoming a permanent running function. We bind that permission and its hash in the release, require the returned identity to equal `glm-5.3`, and expose no standing endpoint.¹

Routes were chosen before each scored block and automatic fallback is disabled. Most endpoints use an exact OpenRouter route; Qwen 3.8 Max uses Alibaba, Fable 5 uses Google Vertex, GLM 5.3 uses the finite Z.ai route above, and selected DeepSeek blocks use an exact GMI-Cloud or DeepInfra route. When transport validation required a replacement, the complete model block was rerun under one new route; successful cells from superseded blocks are not pooled. The manifest binds the requested model, returned identity contract, endpoint tag, provider, supported parameters, and execution contract for both panels. The exact per-model routes appear in Appendix A. For endpoints that advertise a reasoning-effort control, hidden reasoning is excluded and effort is fixed to the lowest transport-stable setting recorded in the exact analysis plan. Endpoints without that control use their provider-fixed behavior. This prevents hidden deliberation from silently consuming the answer budget while leaving the scored response format identical across routes.

¹<https://docs.z.ai/guides/llm/glm-5.3>

3.3 Score-blind complete-core construction

The two task compilers each produce 160 candidates per family. After collection, the release re-scores response syntax with one versioned parser and constructs a validity matrix. Within every panel-family stratum it selects the first 89 eligible task IDs under a fixed SHA-256 ordering. A task is eligible only when all 27 models completed it and produced a parseable three-item set. The selection code does not load the chosen ingredients or their scores until the task IDs are fixed. This yields a balanced common core without favoring any model or post-selecting easy tasks by observed score.

3.4 Uncertainty and multiplicity

All inferential choices are bound in the content-addressed analysis plan.

- **Scores.** 50000 bootstrap samples are drawn over ingredient anchors; tasks that share an anchor across panels always move together. Each resample is reduced to six panel-family means and then equally weighted. We report pointwise percentile intervals and a simultaneous max- t band across all 27 model scores.
- **Pairs.** Every one of the 351 model pairs uses the same 534 tasks. Two-sided anchor-cluster sign-flip tests use 100000 Monte Carlo draws and Holm correction at familywise $\alpha = .05$. We also report paired mean differences, bootstrap intervals, and Cohen’s d_z .
- **Chance.** Each model is compared with the exact taskwise mean over all 56 portfolios. The 27 paired sign-flip tests form a separate Holm family.
- **Ranks.** Point ranks are accompanied by bootstrap rank intervals and contiguous statistical groups. Models that are not significantly separated are not presented as a resolved total order.

We add a retrospective precision diagnostic that does not alter the confirmatory inference. Within each of the six family-panel strata, we draw 5,000 score-blind subsets without replacement at total task counts 30, 60, 90, 150, and 270, then compare each point order with the complete 534-task order. We report rank correlation, top-five overlap, point-leader preservation, pair-order agreement, and score error. A crossed model-by-task decomposition also reports a descriptive generalizability coefficient for relative endpoint comparisons. It treats the release as a

crossed design; it does not turn these fixed endpoints or tasks into random population samples.

Because complete-core eligibility depends on the full endpoint roster, we also run four post-collection sensitivity checks. First, each endpoint is omitted in turn and the six task strata are reselected under the original hash order. Second, each selected stratum is compared with its 160-task candidate pool on five score-map properties, the mean score of the other 26 endpoints, and ingredient category. We compare the 42 observed differences with 20,000 random subsets per stratum and apply Holm correction. Third, we recompute the leaderboard using chance-adjusted gain, action percentile, and exact-optimum rate. Finally, we enumerate all one-percentage-point family weights between 0.20 and 0.50. These diagnostics test whether one roster, task filter, score formula, or moderate family weighting determines the reported order; they do not replace the prespecified analysis.

We also test whether the primary ordering depends on the embedding behind the reward map. This analysis is post hoc. It keeps all 534 prompts, candidate sets, constraints, and 14,418 observed model selections fixed, then recomputes every 56-action map with Epicure-Cooc, Epicure-Core, and Epicure-Chem at immutable Hugging Face revisions [16]. All ingredients used by the benchmark occur in each public vocabulary. We compare task-map and aggregate model orders with the primary map and use 20,000 panel-by-family stratified anchor bootstraps for descriptive intervals. Because the original runtime selected the candidates, this tests reward-map dependence conditional on the released tasks; it is not a newly compiled public-checkpoint benchmark.

Finally, we test the substitution geometry of the public checkpoints against Recipe1MSubs [5], whose standardized test split contains substitutions extracted from recipe-user comments. The analysis protocol is hash-bound before checkpoint scores are inspected. We use exact token matches only, deduplicate directed source-target pairs, and rank each observed target by source-candidate cosine similarity. The primary comparison is deliberately strict: the target is ranked only against ingredients in its own pre-existing food group, so broad category recognition cannot by itself clear the null. Pair percentiles are averaged within source ingredient and then equally across sources; 50,000 source-cluster bootstrap draws form 95% intervals, with Holm correction across the three checkpoint tests. A sensitivity set retains only directed pairs absent from the Recipe1MSubs training split. The labels are external to Epicure training, but the underlying recipes derive from Recipe1M, which is upstream of part of Epicure’s corpus; this is label-independent convergent validation, not corpus-independent validation.

The design has 534 independent anchor clusters and 534 paired cells per contrast. Statistical claims are read directly from simultaneous intervals and Holm-adjusted tests; point-rank differences without corrected support are shown as unresolved.

4 Results

4.1 A statistically resolved FlavourBench leaderboard

All 27 endpoints receive a score on the same 534 tasks. Grok 4.6 has the largest point estimate, 65.1, with simultaneous 95% interval [61.0, 69.2]. Across all 351 prespecified paired contrasts, 101 remain significant after Holm correction. A point rank is therefore a compact summary, not a claim that every adjacent model is distinguishable. Figure 2 shows the score scale, simultaneous intervals, and rank groups; Table 3 gives exact values.

The ranked matrix is rectangular: every row contains 534 valid, parser-confirmed responses, so no endpoint is advantaged by answering a different subset. Provider failures and filtered calls remain available in the full execution ledger, but none appears as a zero-valued culinary decision and no incomplete row enters this leaderboard.

4.2 Does the ordering replicate on a second task panel?

The two panel-specific scores use the same frozen metric and 89 tasks per family, but disjoint task IDs. Their model-level Pearson correlation is 0.89 and their rank correlation is 0.80. These are descriptive stability diagnostics, not a model-selection rule; the primary leaderboard and its uncertainty use both panels and cluster shared ingredient anchors. Figure 3 exposes every model rather than reducing replication to a single coefficient.

4.3 How much task evidence is enough for a stable score?

The crossed design’s relative-decision generalizability coefficient is 0.936 at 534 tasks; the same variance model estimates 329 balanced tasks for 0.90. In 5,000 stratified half-size subsamples (270 tasks), the median rank correlation with the complete point order is 0.952 (empirical 95% range 0.897–0.980) and median top-five overlap is 80%. The complete point leader is retained in only 46.1% of those subsets. The first two results show that 534 tasks support a reliable global score; the last shows why a narrow claim about the single point leader would be stronger than the data allow.

4.4 Which pairwise conclusions survive correction?

The score is continuous, and each model pair is compared on the same 534 tasks rather than on a small number of arena votes. Figure 5 visualizes the multiplicity-controlled results. Grey cells are scientifically useful: they mark pairs for which the shared task evidence does not support a directional claim at familywise $\alpha = .05$. This is different from declaring the two models equal.

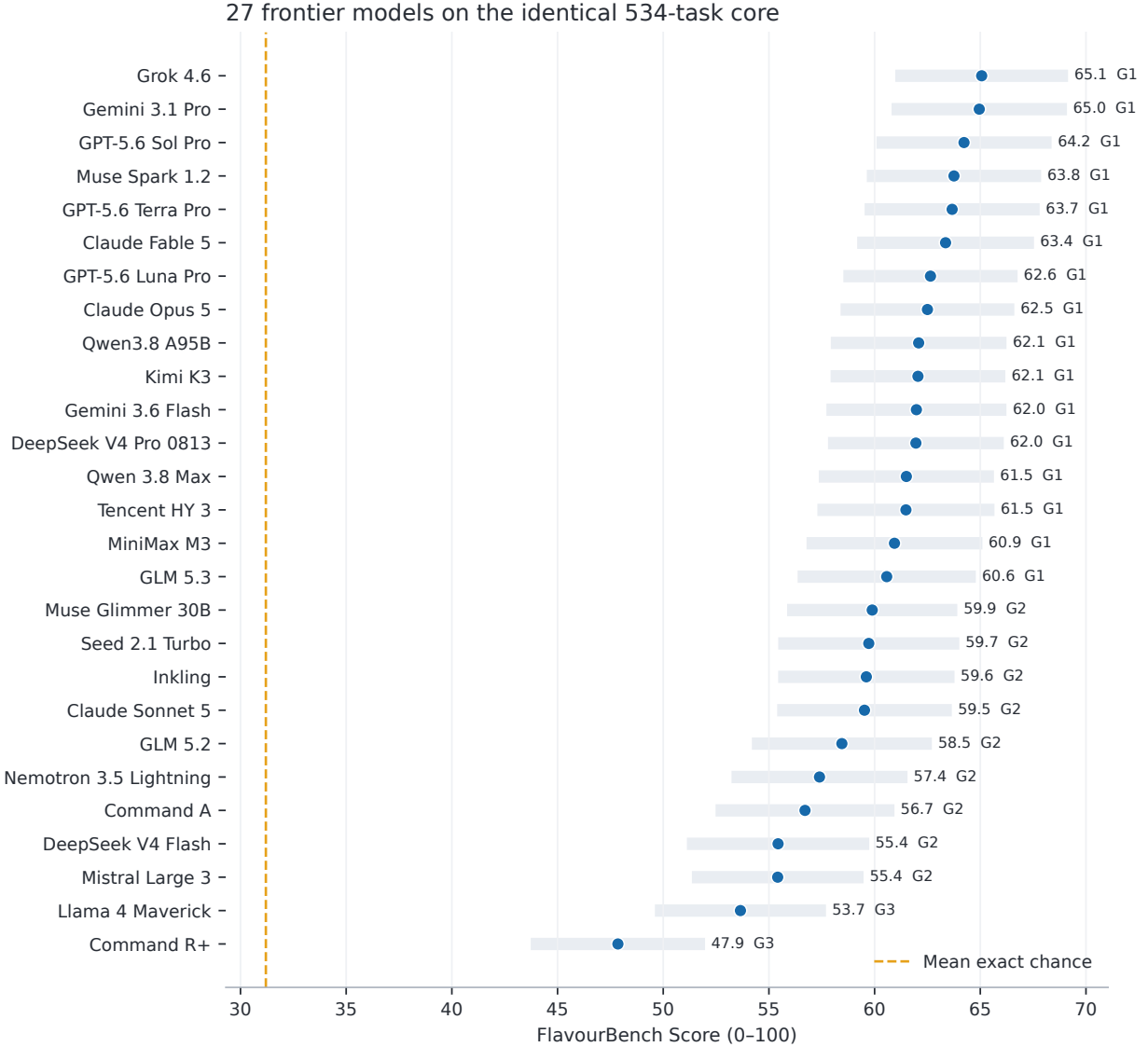


Figure 2: FlavourBench Score with simultaneous 95% max- t intervals. The letter-free group numbers are inferential tiers: a point rank is shown for navigation, while models inside an unresolved group should not be read as a statistically established ordering.

4.5 Does the task filter or score formula determine the order?

The endpoint whose candidate-panel validity most constrained the common-core filter was Claude Fable 5. Omitting it and reapplying the original task-selection hash retains 71.0% of the official tasks. Among the remaining 26 endpoints, the new scores differ by 0.65 points on average and at most 1.37 points; rank correlation is 0.955, 92.9% of pair directions agree, and the point leader is unchanged. The other omissions alter at most three tasks per stratum.

The selected and unselected candidate tasks differ by at most 0.27 pooled standard deviations across the six numerical diagnostics. None of the 42 numerical or category comparisons is resolved after Holm correction against the random-subset reference. The score itself is not the

only summary that recovers the broad order: Table 4 shows rank correlations from 0.938 to 0.985 and pair-order agreement of at least 89.2%. Across 696 moderate family-weight combinations, rank correlation with equal weighting never falls below 0.980. The point leader can change between Grok 4.6 and Gemini 3.1 Pro, which reinforces the primary analysis: the global ordering is stable, but the data do not identify a unique best endpoint.

4.6 Is the ordering tied to one Epicure embedding?

The three public checkpoints change the individual reward maps substantially. Median task-level rank correlation with the primary map ranges from 0.660 to 0.752, and the exact optimum agrees on 27.7% to 38.4% of tasks.

Rank	Model	FB Score	simultaneous 95% CI	rank 95% CI	group
1	Grok 4.6	65.1	[61.0, 69.2]	[1, 5]	1
2	Gemini 3.1 Pro	65.0	[60.8, 69.1]	[1, 6]	1
3	GPT-5.6 Sol Pro	64.2	[60.1, 68.4]	[1, 8]	1
4	Muse Spark 1.2	63.8	[59.6, 67.9]	[1, 10]	1
5	GPT-5.6 Terra Pro	63.7	[59.5, 67.8]	[1, 11]	1
6	Claude Fable 5	63.4	[59.2, 67.5]	[2, 13]	1
7	GPT-5.6 Luna Pro	62.6	[58.5, 66.8]	[4, 14]	1
8	Claude Opus 5	62.5	[58.4, 66.6]	[3, 15]	1
9	Qwen3.8 A95B	62.1	[57.9, 66.2]	[4, 17]	1
10	Kimi K3	62.1	[57.9, 66.2]	[5, 16]	1
11	Gemini 3.6 Flash	62.0	[57.7, 66.2]	[4, 17]	1
12	DeepSeek V4 Pro 0813	62.0	[57.8, 66.1]	[5, 17]	1
13	Qwen 3.8 Max	61.5	[57.4, 65.6]	[7, 18]	1
14	Tencent HY 3	61.5	[57.3, 65.7]	[6, 19]	1
15	MiniMax M3	60.9	[56.8, 65.1]	[8, 20]	1
16	GLM 5.3	60.6	[56.4, 64.8]	[8, 20]	1
17	Muse Glimmer 30B	59.9	[55.9, 63.9]	[12, 21]	2
18	Seed 2.1 Turbo	59.7	[55.4, 64.0]	[12, 22]	2
19	Inkling	59.6	[55.4, 63.8]	[12, 22]	2
20	Claude Sonnet 5	59.5	[55.4, 63.7]	[13, 22]	2
21	GLM 5.2	58.5	[54.2, 62.7]	[16, 23]	2
22	Nemotron 3.5 Lightning	57.4	[53.2, 61.6]	[19, 25]	2
23	Command A	56.7	[52.5, 60.9]	[20, 25]	2
24	DeepSeek V4 Flash	55.4	[51.1, 59.7]	[22, 26]	2
25	Mistral Large 3	55.4	[51.4, 59.5]	[22, 26]	2
26	Llama 4 Maverick	53.7	[49.6, 57.7]	[24, 26]	3
27	Command R+	47.9	[43.7, 52.0]	[27, 27]	3

Table 3: The complete automated leaderboard. Every score and every pairwise comparison uses the identical 534-task core. Rank intervals come from the anchor-cluster bootstrap; groups summarize the Holm-controlled pairwise graph.

Score summary	Rank ρ	Pair order	Point leader
FlavourBench Score	1.000	100.0%	Grok 4.6
Chance-adjusted gain	0.985	96.0%	Gemini 3.1 Pro Preview
Action percentile	0.969	93.4%	Grok 4.6
Exact-optimum rate	0.938	89.2%	Grok 4.6

Table 4: Sensitivity to three alternative summaries of the same 534 decisions. Correlation and pair order are computed against the primary FlavourBench Score. These are descriptive robustness checks, not additional confirmatory rankings.

Reward map	Task-map ρ	Model-rank ρ	Pair order	Point leader
Epicure-Cooc	0.752	0.957	91.7%	Grok 4.6
Epicure-Core	0.672	0.915	88.0%	Grok 4.6
Epicure-Chem	0.660	0.903	86.9%	Grok 4.6

Table 5: Post-hoc replacement of the reward map on the fixed 534 tasks and fixed model selections. Task-map ρ is the median correlation across each task’s 56 portfolios. Model rank and pair order compare the aggregate leaderboard with the primary map.

Despite those local changes, the aggregate model order is similar: model-rank correlation ranges from 0.903 to 0.957, and 86.9% to 91.7% of all model-pair directions agree. Grok 4.6 has the largest point estimate under Cooc, Core, and Chem. In the stratified bootstrap, median model-rank correlation ranges from 0.875 to 0.930. The same sampled point leader appears under the primary and replacement map in only 38.9% to 53.8% of draws, so this result supports the broad ordering rather than a definitive winner.

Checkpoint	Percentile [95% CI]	Novel pairs	Hit@10
Cooc	0.806 [0.788, 0.824]	0.754	0.133
Core	0.800 [0.781, 0.819]	0.735	0.172
Chem	0.780 [0.761, 0.798]	0.718	0.155

Table 6: Label-independent validation on unique mapped Recipe1MSubs test pairs. Percentile is the equal-source rank of the observed target among same-food-group candidates; novel pairs are absent from the Recipe1MSubs training split. Hit@10 uses the full 1,789-candidate vocabulary.

4.7 Do public Epicure checkpoints recover observed substitutions?

Exact matching maps 3,282 of 10,747 Recipe1MSubs test events to the public 1,790-ingredient vocabulary, yielding 1,469 unique directed pairs over 357 source ingredients. Cooc, Core, and Chem place the observed target at equal-source within-food-group rank percentiles 0.806, 0.800, and 0.780, respectively; their source-clustered 95% intervals are [0.788, 0.824], [0.781, 0.819], and [0.761, 0.798]. All three one-sided tests reject the 0.5 chance null after Holm correction ($p_{\text{Holm}} < 10^{-4}$). On the 594 directed pairs absent from the Recipe1MSubs training split, the corresponding percentiles remain 0.754, 0.735, and 0.718. Full-vocabulary Hit@10 is 0.133–0.172, compared with an analytic random baseline of 0.0056.

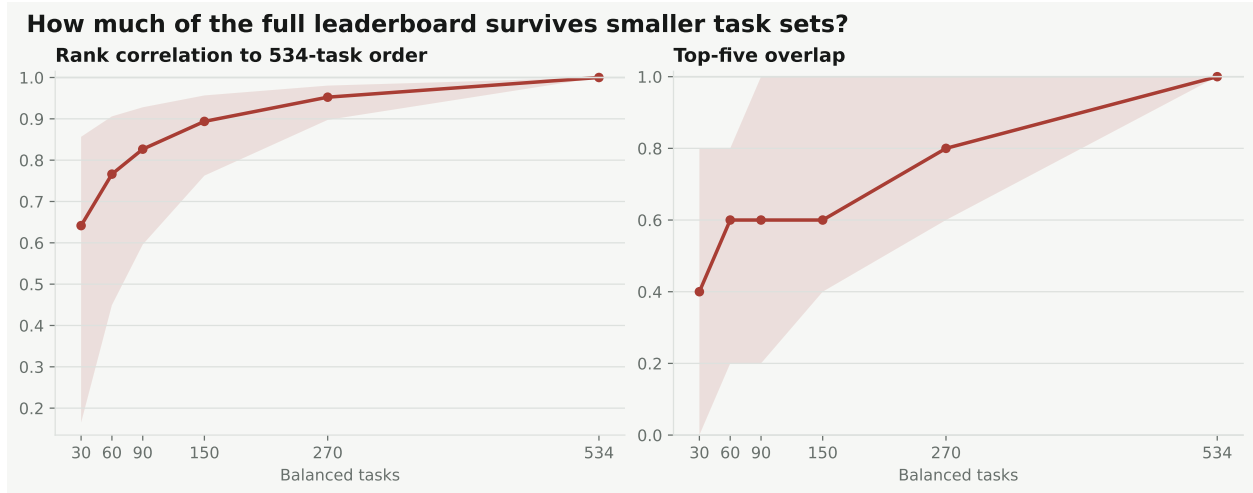


Figure 4: Stability of smaller, score-blind task subsets relative to the complete 534-task point order. Lines are medians across family-panel stratified samples; bands are empirical 2.5th to 97.5th percentiles. The 534-task endpoint is identical to the reference by construction. This is a precision diagnostic, not a post-hoc power calculation.

Family	Prompt	Higher-ranked selection	Lower-ranked selection
Substitution	Select three alternatives to ‘boursin cheese’ that best preserve its dairy role, regional context, and mutual portfolio coherence.	caciocavallo, fromage blanc, grana padano (100)	fromage blanc, quail egg, goose egg (15)
Pairing	Select the three-ingredient bundle with the strongest learned pairing to ‘sweetbread’ and the best internal coherence.	parsnip, cipollini onion, porcini mushroom (92)	parsnip, pickled onion, peppadew pepper (0)
Constraints	For a dish centered on ‘potato’, select three vegetarian complements with NOVA processing level at most 2; among valid portfolios, maximize learned pairing and coherence.	parsley, black pepper, thyme (100)	cheese, bay leaf, parsley (0)

Table 7: Three common-core response examples. Parentheses give the fixed task score. The complete prompts and unabridged responses are released in machine-readable form.

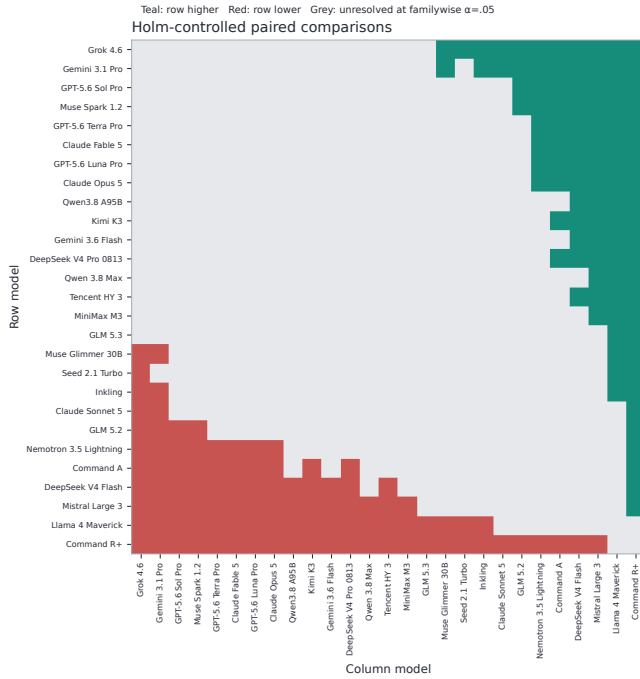


Figure 5: All paired model contrasts after Holm correction. Teal means the row model scores higher; red means lower; grey is unresolved.

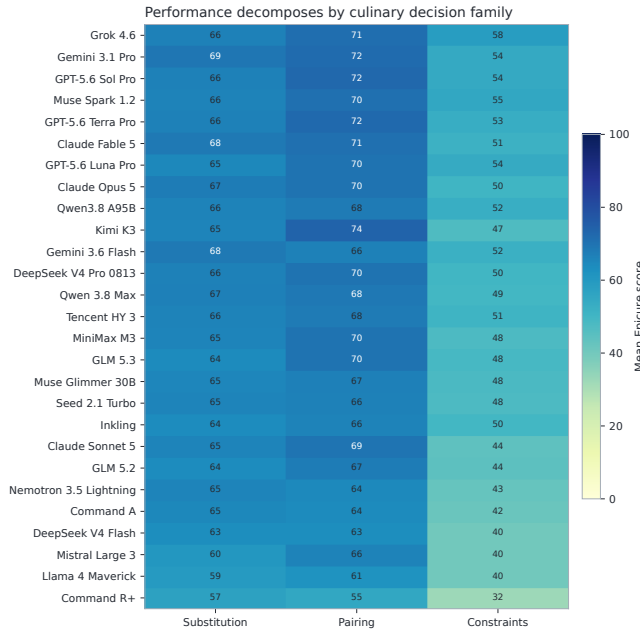


Figure 6: Mean score by family on the common core. Each family contributes 178 tasks per model. Values share the 0–100 scale but retain family-specific decision semantics.

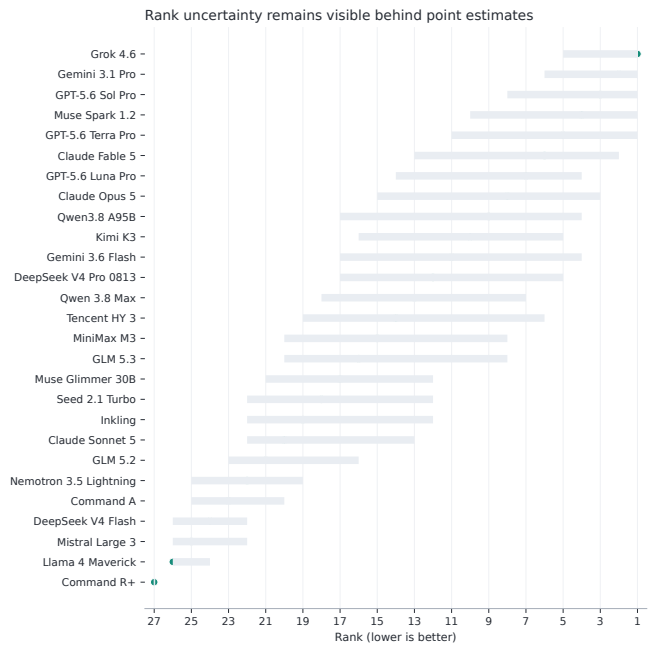


Figure 7: Point rank and 95% bootstrap rank interval for every endpoint. Narrow intervals imply a stable location; overlapping intervals mark unresolved local order.

5 Reward transfer beyond format learning

The exhaustive maps can supervise a model as well as score it. We test whether EPICURE supervision transfers to unseen maps beyond learning the required answer syntax. The machine-readable protocol and evaluator were released before optimization outcomes were generated. The study pins Qwen3-0.6B at one repository revision [22], uses 270 training and 72 validation tasks, and reserves 84 primary-transfer tasks. Ingredient anchors, candidate sets, and prompt hashes are disjoint across those splits; all 534 leaderboard maps form a declared secondary replication.

Three seeds receive three epochs of all-linear LoRA SFT [7] with rank 16, effective batch size 16, learning rate 10^{-4} , completion-only loss, and the final checkpoint without validation-based selection. The treatment completion is each task’s EPICURE optimum. A naive base-versus-SFT comparison would confound reward learning with practice emitting `FINAL_SELECTION`. We therefore rotate the same A–H portfolios onto different prompts within each family and source panel. Treatment and control have identical prompts, row counts, completion lengths, and portfolio-label histograms, but the control contains no accidental task optimum. Evaluation uses greedy decoding with thinking disabled and a 64-token limit. Unparseable completions remain in the score at zero.

The preregistered estimand is the equal-family treatment-minus-control score difference. A 50,000-draw crossed bootstrap resamples matched training seeds and ingredient anchors within the six family–panel strata. A two-sided 100,000-draw sign-flip test operates on per-anchor effects averaged over seeds. The interval therefore includes empirical seed and anchor variation, whereas the p -value tests held-out-anchor effects conditional on the three realized seed pairs; it is not a population-level test over retraining randomness. Three points is the preregistered practical threshold. There is one primary contrast, so no multiplicity correction is required.

On the 84 anchor-disjoint tasks, EPICURE SFT scores 44.20 versus 30.90 for the format control, a 13.30-point gain (95% CI 6.52–20.29, $p = 1.70 \times 10^{-4}$). All three primary seed effects are positive. Both trained arms parse 100.00% of responses, while the base parses 89.29%. Format training changes the base score by only 0.90 points (95% CI -4.91–6.85, $p = 0.790$). The secondary public-map replication estimates a 11.73-point effect (95% CI 8.98–14.54, $p = 1.00 \times 10^{-5}$), with all three replication seed effects positive.

The control is empirically consequential. On the public maps it improves on the base by 3.05 points (95% CI 0.46–5.68, $p = 0.021$), consistent with removing parse failures. A treatment–base comparison would fold that syntax gain into the reward effect. Treatment–control differences are positive in all six evaluation-by-family cells and largest for constraints in both evaluations; these family rows are descriptive rather than separately tested.

This is evidence for learning the released reward function rather than answer syntax. It is not evidence of better human taste, cooked-food quality, general language-model ability, reinforcement-learning improvement, or transfer outside this environment. The same maps expose SFT, preference, and scalar-reward interfaces for larger follow-up studies, but only SFT is tested here.

Evaluation	Base	Format control	Epicure SFT	Δ [95% CI]	p
Anchor-disjoint transfer ($n = 84$)	29.99	30.90	44.20	+13.30 [6.52, 20.29]	1.70×10^{-4}
Public-map replication ($n = 534$)	28.56	31.60	43.33	+11.73 [8.98, 14.54]	1.00×10^{-5}

Table 8: Reward transfer. Scores average three training seeds except the unmodified base. Δ is Epicure SFT minus format control; the public-map row is a declared secondary replication opened only after the primary analysis was sealed.

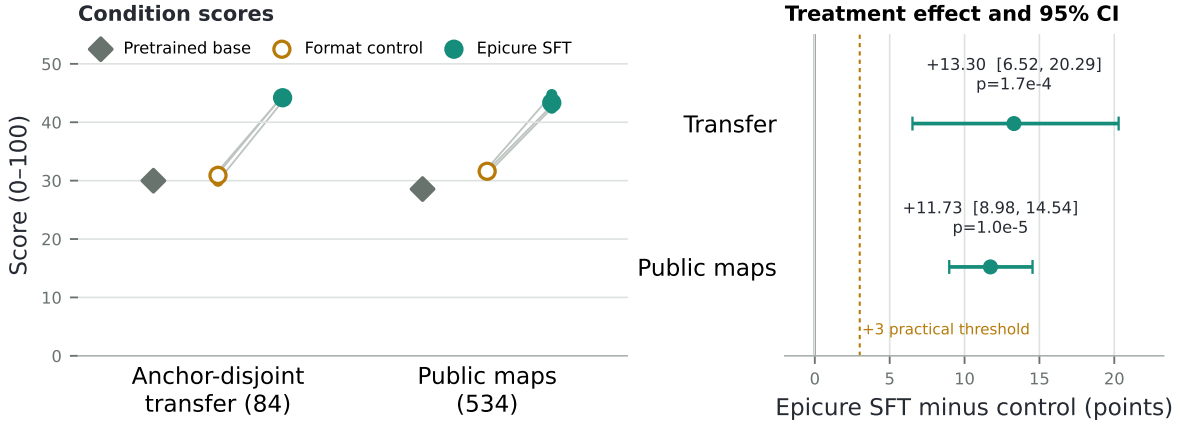


Figure 8: Held-out scores and matched treatment effects. Small points are training seeds; connecting lines pair control and treatment adapters initialized with the same seed. The right panel reports the prespecified crossed-bootstrap interval. The dashed line is the three-point practical threshold, not a null hypothesis.

6 Reproducibility

The content-addressed release ([source](#), [dataset](#), and [leaderboard](#)) includes:

- both exact content-addressed task sets (534 scheduled tasks and 534 unique anchors), with prompts, candidates, all 56 portfolio scores, oracle provenance, task strata, and task-set digests;
- the 27-model manifest, exact provider route per model, fallback policy, supported parameters, endpoint contract, and route digest;
- all response records used to construct the 14418-cell ranked matrix, including raw text, completion state, usage, latency, parser output, task score, and digest;
- all content-addressed provider-attempt events referenced by those responses, including normalized and native finish reasons used to distinguish refusals from ordinary completions;
- the frozen joint analysis plan, anchor-cluster contract, seeds, 50,000 bootstrap and 100,000 sign-flip replicates, all 351 pairwise rows, cross-panel diagnostics, and the final statistical release;
- the three pinned public-checkpoint reward maps, task-level agreement records, alternative leaderboards, and 20,000-draw scorer-sensitivity analysis;
- the hash-bound Recipe1MSubs protocol, aggregate substitution-validation artifact, source- clustered intervals, and a builder that retrieves official raw files by immutable hash without redistributing them; and

- the frozen reward-transfer plan, format-matched control data, six training manifests, raw generations on both held-out evaluations, resampling outputs, and adapter hashes; and
- a verifier and asset builder that reconstruct the leaderboard CSV, pairwise CSV, LaTeX tables, and vector figures from local files without provider access.

Responses are written once under a model slot and cell identifier. Existing valid cells are skipped on resume; conflicting artifacts fail closed. Provider fallback is disabled, so the route shown in the manifest is the route measured. The task maps and analysis plan were frozen before the clean primary lineage; transport-calibration responses are retained separately and never enter the leaderboard.

7 Related work

General-purpose evaluation spans broad suites such as HELM [13], difficult knowledge tests such as MMLU-Pro and GPQA [19, 17], live contamination-resistant sets [20, 9], and human preference systems such as Chatbot Arena [2]. Model judges increase scale but introduce their own bias and calibration problems [24]. FLAVOURBENCH instead takes the executable- environment route used by software and agent benchmarks: SWE-bench runs repository tests [10], BFCL evaluates function calls [14], and τ -bench checks interaction outcomes against database state [23]. RewardBench evaluates learned reward models on chosen-rejected pairs [12]. FLAVOURBENCH uses a versioned program as its reward source and asks whether direct supervision from that program changes decisions on unseen task states after controlling for response syntax.

Culinary datasets and benchmarks cover ingredient networks, recipe understanding, planning, nutrition, and food knowledge [1, 6, 18, 5, 3, 21, 8, 4, 11]. Our focus is narrower and complementary: common-task frontier model measurement against a released culinary decision environment, with exhaustive partial-credit maps and paired inference.

8 Limitations

FlavourBench Score measures agreement with a released reward map, not universal human taste. The primary runtime is reproducible by content hash, but its original training run and seed are unrecovered; the published Epicure sibling papers do not validate that exact artifact. Rescoring the fixed decisions with Cooc, Core, and Chem recovers the broad model order, but the original runtime still chose the candidate sets. Recipe1MSubs supplies held-out human-observed substitution labels for the public checkpoints, but its recipes share Recipe1M ancestry with part of Epicure’s corpus; it neither validates the unrecovered primary runtime nor tests the pairing and constraint components. None of these checks supplies human sensory or cooking-outcome validation.

Within-task min-max normalization makes the 0–100 scale comparable while discarding absolute raw-utility magnitudes. The tasks evaluate constrained ingredient selection rather than full recipe generation, sensory execution, safety advice, or long-horizon kitchen planning. The common-core requirement removes differential missingness from ranking, but it restricts the estimand to tasks completed by every endpoint. The roster analysis measures the observed effect of this filter; it cannot rule out dependence on task properties we did not record. Results also bind particular model routes and collection dates. The dense reward surface supports learning experiments, but the transfer study uses LoRA SFT of one 0.6B checkpoint over three seeds. It does not compare training algorithms or model scales. The public-map result reuses the same adapters, so it is an independent task replication rather than an independent training replication. Neither transfer evaluation tests human cooking outcomes.

9 Conclusion

FLAVOURBENCH measures culinary decisions without asking one language model to grade another. A content-addressed reward map enumerates every candidate decision, a 27-model complete panel produces one continuous score, and paired uncertainty distinguishes resolved differences from apparent rank order. Controlled SFT then shows that the same maps can alter decisions on unseen anchors beyond format practice. Under the released culinary environment, every leaderboard score, training contrast, interval, table, and figure is reconstructable from public artifacts.

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A Exact execution routes

Model	Backend	Panel 1 route	Panel 2 route
Grok 4.6	openrouter	xai/zdr	xai/zdr
Gemini 3.1 Pro	openrouter	google-ai-studio	google-ai-studio
GPT-5.6 Sol Pro	openrouter	openai/flex	openai/flex
Muse Spark 1.2	openrouter	meta	meta
GPT-5.6 Terra Pro	openrouter	openai/flex	openai/flex
Claude Fable 5	openrouter	google-vertex/global	google-vertex/global
GPT-5.6 Luna Pro	openrouter	openai/flex	openai
Claude Opus 5	openrouter	azure/us	azure/us
Qwen3.8 A95B	openrouter	alibaba	alibaba
Kimi K3	openrouter	morph	morph
Gemini 3.6 Flash	openrouter	google-vertex/us	google-vertex/us
DeepSeek V4 Pro 0813	openrouter	gmcloud/fp8	gmcloud/fp8
Qwen 3.8 Max	openrouter	alibaba	alibaba
Tencent HY 3	openrouter	gmcloud/bf16	gmcloud/bf16
MiniMax M3	openrouter	venice/fp8	venice/fp8
GLM 5.3	zai_coding_direct	zai-coding-plan-direct	zai-coding-plan-direct
Muse Glimmer 30B	openrouter	fireworks	fireworks
Seed 2.1 Turbo	openrouter	seed/fp8	seed/fp8
Inkling	openrouter	deepinfra/fp8	deepinfra/fp8
Claude Sonnet 5	openrouter	amazon-bedrock/claude-on-aws	amazon-bedrock/claude-on-aws
GLM 5.2	openrouter	coreweave/fp4	coreweave/fp4
Nemotron 3.5 Lightning	openrouter	venice/fp4	venice/fp4
Command A	openrouter	cohere	cohere
DeepSeek V4 Flash	openrouter	deepinfra/fp4	deepinfra/fp8
Mistral Large 3	openrouter	mistral	mistral
Llama 4 Maverick	openrouter	digitalocean	digitalocean
Command R+	openrouter	cohere	cohere

Table 9: Frozen execution routes by panel. Automatic fallback is disabled; route differences indicate complete, score-blind model-block replacements rather than cell pooling.

B Per-family scores

Model	Substitution	Pairing	Constraints
Grok 4.6	66.1	70.9	58.2
Gemini 3.1 Pro	69.0	71.6	54.2
GPT-5.6 Sol Pro	66.2	72.5	54.1
Muse Spark 1.2	65.9	70.3	55.0
GPT-5.6 Terra Pro	66.1	72.1	52.8
Claude Fable 5	68.2	70.6	51.3
GPT-5.6 Luna Pro	64.5	69.7	53.7
Claude Opus 5	67.2	69.9	50.4
Qwen3.8 A95B	66.0	67.8	52.5
Kimi K3	65.5	73.7	47.0
Gemini 3.6 Flash	68.1	66.2	51.6
DeepSeek V4 Pro 0813	66.4	69.5	49.9
Qwen 3.8 Max	66.6	68.5	49.5
Tencent HY 3	66.0	67.9	50.5
MiniMax M3	65.4	69.9	47.6
GLM 5.3	63.7	69.6	48.4
Muse Glimmer 30B	65.1	66.8	47.8
Seed 2.1 Turbo	65.3	66.2	47.6
Inkling	63.6	65.8	49.5
Claude Sonnet 5	65.3	69.2	44.1
GLM 5.2	63.6	67.2	44.5
Nemotron 3.5 Lightning	65.3	64.3	42.5
Command A	64.7	63.9	41.6
DeepSeek V4 Flash	63.1	63.2	40.1
Mistral Large 3	60.4	65.8	40.0
Llama 4 Maverick	59.4	61.4	40.2
Command R+	56.7	54.9	32.0

Table 10: Mean score in each 178-task family. Every entry uses the same tasks for every model; the equal-family mean is the FlavourBench Score.

C Protocol constants

Quantity	Frozen value
Models	27
Collection panels	2
Primary tasks per model	534
Unique anchor clusters	534
Tasks per family	178
Candidate ingredients	8
Selected ingredients	3
Scored portfolios per task	56
Complete response cells	14418
Bootstrap replicates	50000
Sign-flip replicates	100000
Pairwise hypotheses	351
Shared cells per pair	534
Common-core rule	Status/parser only; score-blind

Table 11: Core design and inference constants fixed before clean primary execution.

D Task-selection and score sensitivity

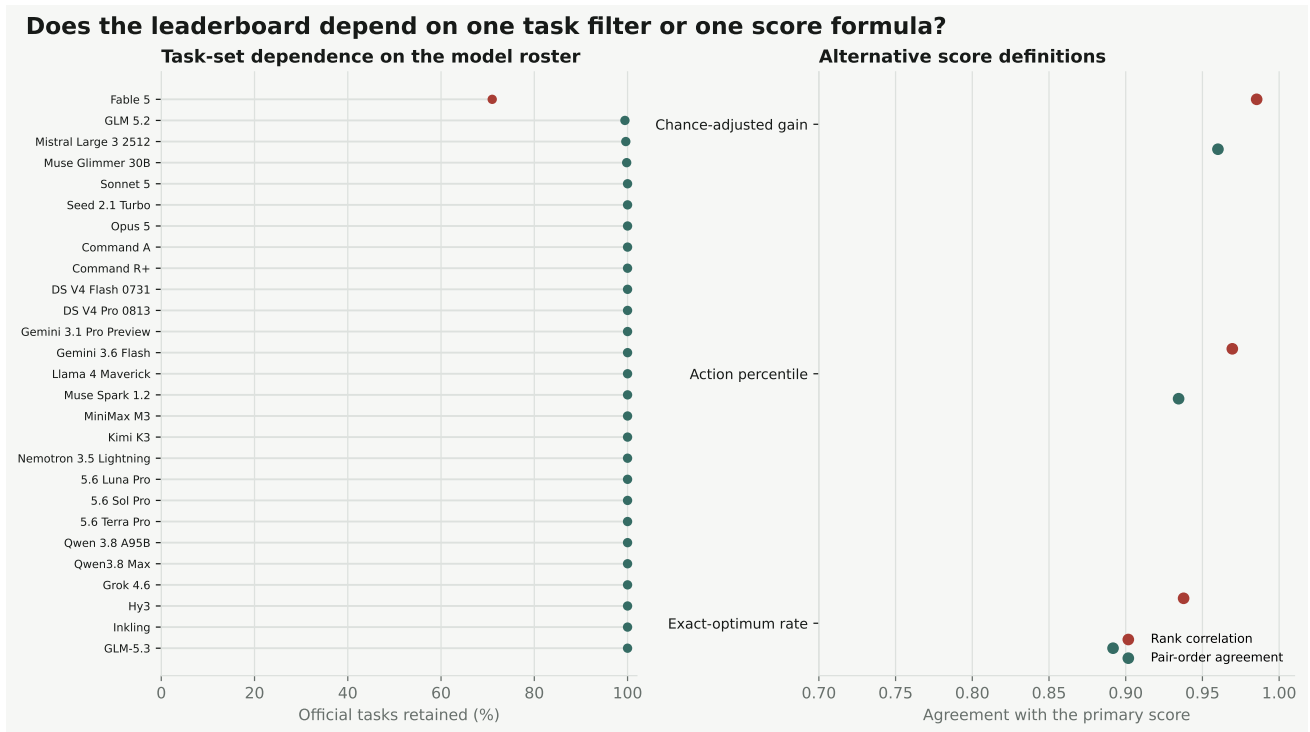


Figure 9: Left: fraction of the official 534 tasks retained after omitting each endpoint and rerunning the frozen hash selection. Right: agreement between the primary score and three alternative summaries of the same decisions. The analysis is post-collection and does not alter the confirmatory release.

E Public-checkpoint scorer sensitivity

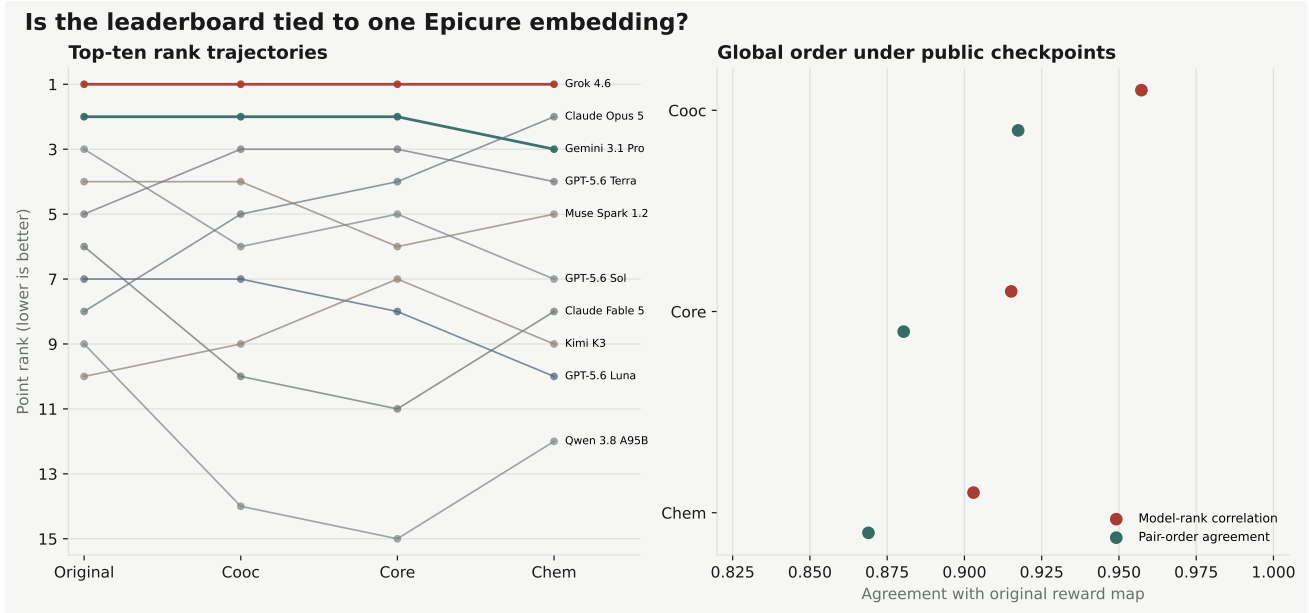


Figure 10: Left: point-rank trajectories for the primary top ten when the same observed selections are scored with three public Epicure checkpoints. Right: model-rank correlation and model-pair direction agreement with the primary reward map. The analysis keeps the released tasks fixed and is post hoc.

F External substitution validation

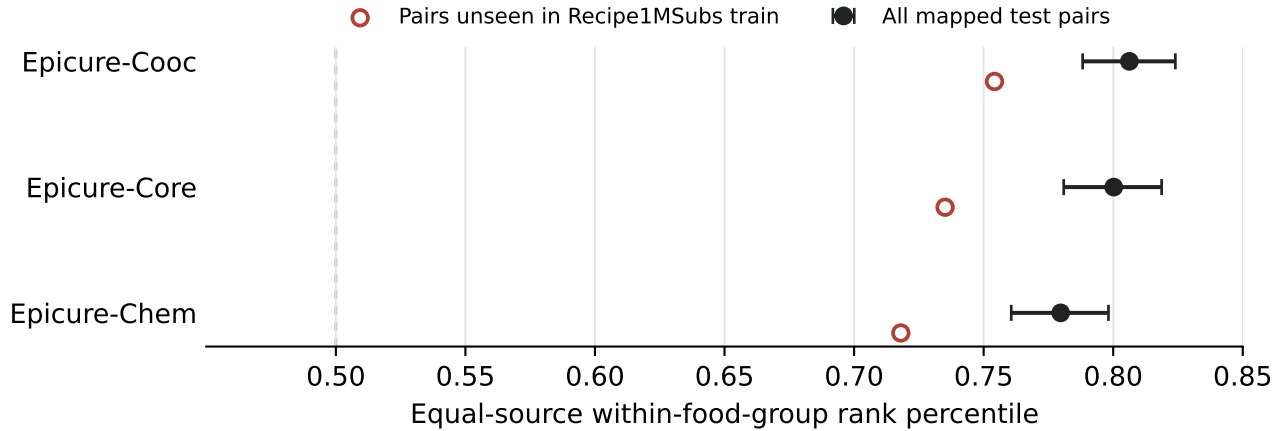


Figure 11: Observed Recipe1MSubs targets ranked against same-food-group alternatives. Filled points and source-clustered 95% intervals use all 1,469 unique mapped test pairs; open points retain only the 594 directed pairs absent from Recipe1MSubs training. The dashed line is the 0.5 chance percentile.

G Reward-transfer protocol and diagnostics

Component	Frozen value
Base	Qwen3-0.6B, revision <code>c1899de289a04d1...</code>
Conditions	unmodified base; format-control SFT; Epicure-optimum SFT
Train / validation / transfer	270 / 72 / 84 anchor-disjoint tasks
Training seeds	20260824, 20260825, 20260826
LoRA	rank 16; alpha 32; dropout .05; all linear layers
Optimization	3 epochs; AdamW; LR 10^{-4} ; effective batch 16
Checkpoint rule	final adapter only; no validation selection
Decoding	greedy; thinking disabled; 64 new tokens
Primary score	equal family and panel; unparseable responses score zero
Inference	50,000 crossed bootstraps; 100,000 anchor sign flips

Table 12: Frozen reward-transfer protocol. Validation loss is monitored but never selects a checkpoint.

Evaluation	Family	Base	Format control	Epicure SFT	Δ
Transfer	Substitution	57.93	41.62	52.79	+11.16
Transfer	Pairing	28.66	46.34	53.56	+7.23
Transfer	Constraint	3.39	4.73	26.24	+21.51
Public	Substitution	47.63	47.43	52.39	+4.96
Public	Pairing	34.36	43.59	53.62	+10.02
Public	Constraint	3.69	3.78	23.99	+20.21

Table 13: Descriptive scores by task family. Δ is Epicure SFT minus format control; the family rows were not separate confirmatory hypotheses.

Seed	Anchor-disjoint transfer			Public-map replication		
	Control	Epicure	Δ	Control	Epicure	Δ
20260824	31.61	44.47	+12.86	32.06	44.74	+12.69
20260825	31.19	44.53	+13.34	31.10	42.40	+11.30
20260826	29.89	43.59	+13.70	31.65	42.85	+11.20

Table 14: Matched training-seed scores. Each Δ compares adapters initialized with the same seed.